



Teacher Copy — includes the Answer Key & Teacher Guide. Do not distribute to students.

UNIT 5 • STANDARD 6.GR.2

Volume of Rectangular Prisms

Lesson 5-10 • Teacher Copy

Name: _____

Date: _____

Class: _____



TODAY'S OBJECTIVES

Content Objective: I can find the volume of a rectangular prism, including ones with fractional edge lengths, using $\text{base area} \times \text{height}$.

Language Objective: I can explain my work using the words volume, rectangular prism, dimensions, and base area.



See **Learn It** for the explanation and worked examples, and the **Vocab** tab for the words. These are your notes to fill in and keep.



Fill in each blank as we go. Use the Word Bank to help you.



WORD BANK — FILL EACH BLANK WITH THE BEST WORD

Volume of Rectangular Prisms

Volume

Rectangular prism

Cubic units

Dimensions

Net

Base area



Tap any word to see what it means and a picture.

Level 3: try the blanks from memory first, then check the bank.

1

The space inside a rectangular prism, found with $V = l \times w \times h$, is the

2

The amount of space inside a three-dimensional solid is its

3

A solid with six rectangular faces is a .

4

The unit used to measure volume is .

5

The measurements of length, width, and height of a solid are its

6 A flat pattern that folds into a solid is a .

7 The area of the bottom face of a prism, multiplied by height to find volume, is the .

Write About the Math

Guided ESOL writing

Use the support level you need. Every level answers the same math question.

1. Understand the Question

EXPLAIN

How many gallons of water does the small tank hold?

Your job: Answer the question. Use math evidence. Explain how the evidence proves your answer.

Responde la pregunta. Usa evidencia matemática. Explica cómo la evidencia demuestra tu respuesta.

2. Plan Your Math Words

Check at least two words you will use.

- ☐ **Volume of Rectangular Prisms** — The space inside a rectangular prism.
Formula: $V = Bh = l \times w \times h$.
Volumen de prismas rectangulares — El espacio dentro de un prisma rectangular.
Fórmula: $V = Bh = l \times w \times h$.

- ☐ **Volume** — How much space is inside a solid shape.
Volumen — Cuánto espacio hay dentro de una figura sólida.

- ☐ **Rectangular prism** — A solid box shape with six flat rectangle sides.
Prisma rectangular — Una figura sólida en forma de caja con seis lados rectangulares.

- ☐ **Cubic units** — The units used to measure space inside, like cubic inches.
Unidades cúbicas — Las unidades para medir el espacio interior, como pulgadas cúbicas.

- ☐ **Dimensions** — How long, how wide, and how tall a shape is.
Dimensiones — Qué tan largo, ancho y alto es una figura.

Say it first: Say your idea to a partner or quietly to yourself before you write.

Di tu idea a un compañero o en voz baja antes de escribir.

The tank's volume is _____ in³ because $V = ___ \times ___ \times ___$. It holds about _____ gallons because $___ \div 231 \approx ___$.

Di tu respuesta primero: Mi respuesta es _____ porque _____.

3. Build Your Explanation

Model the parts: This model shows the parts of an explanation. It does not answer your writing question.

Claim: You still multiply length $\times$ width $\times$ height even when an edge is a fraction or decimal.

Evidence: Students explain the formula stays the same ($V = l \times w \times h = 2 \times 1.5 \times 1 = 3 \text{ ft}^3$) and that a fractional edge just means multiplying with a fraction or decimal.

Reasoning: This evidence connects the definition of volume of rectangular prisms to the mathematical claim.

Start

Most language support

Complete the frames. Use the word bank.

Completa las oraciones. Usa el banco de palabras.

- The tank's volume is ____ in^3 because $V = \text{____} \times \text{____} \times \text{____}$. It holds about ____ gallons because $\text{____} \div 231 \approx \text{____}$.
- My answer is ____.

Mi respuesta es ____.

Build

Some language support

Write two connected sentences. Use because, so, or but.

Escribe dos oraciones conectadas. Usa porque, entonces o pero.

- My claim is ____.
- I know because ____.

Mi afirmación es ____.

Lo sé porque ____.

- So ____.

Entonces ____.

Explain

Light language support

Write a claim, give math evidence, and explain your reasoning.

Escribe una afirmación, da evidencia matemática y explica tu razonamiento.

- **My claim is ____.**
Mi afirmación es ____.
- **My math evidence is ____.**
Mi evidencia matemática es ____.
- **This proves ____ because ____.**
Esto demuestra ____ porque ____.

4. Check Your Explanation

- ☐ I answered the question.
Respondí la pregunta.
- ☐ I used math evidence: numbers, an equation, a model, or a comparison.
Usé evidencia matemática: números, una ecuación, un modelo o una comparación.
- ☐ I explained how the evidence proves my answer.
Explicué cómo la evidencia demuestra mi respuesta.
- ☐ I used at least two math words.
Usé por lo menos dos palabras matemáticas.
- ☐ I reread complete sentences and punctuation.
Volví a leer mis oraciones completas y la puntuación.

Writing Guide — Teacher Copy

The support level changes the amount of language scaffolding, not the mathematical expectation. Look for the same five features in every response.

- I answered the question.
- I used math evidence: numbers, an equation, a model, or a comparison.
- I explained how the evidence proves my answer.
- I used at least two math words.
- I reread complete sentences and punctuation.

Answer Key & Teacher Guide

1. **Notes 1:** Volume of Rectangular Prisms
2. **Notes 2:** Volume
3. **Notes 3:** Rectangular prism
4. **Notes 4:** Cubic units
5. **Notes 5:** Dimensions
6. **Notes 6:** Net
7. **Notes 7:** Base area
8. **Watch for this mistake:** *A common mistake in Volume of Rectangular Prisms is rounding a fractional edge length to the nearest whole number before multiplying, instead of using the exact fraction or decimal. For a box that is $2\text{ ft} \times 1.5\text{ ft} \times 1\text{ ft}$, a student might round 1.5 down to 1 and compute $2 \times 1 \times 1 = 2\text{ ft}^3$, but the correct volume uses the exact edge length: $2 \times 1.5 \times 1 = 3\text{ ft}^3$ — a full cubic foot larger. Before you submit, ask: 'Did I multiply using the exact fractional or decimal edge length, or did I round it off first?'*
9. **Try It 1:** A. Box A, because 50 cubic in is more than 48 cubic in — Box A: $5 \times 5 \times 2 = 50$ cubic in. Box B: $4 \times 4 \times 3 = 48$ cubic in. $50 > 48$, so Box A holds more.
10. **Try It 2:** A. 18 cubic in — $V = 4 \times 3 \times 1.5 = 12 \times 1.5 = 18$ cubic in. A fractional edge does not change the steps.